NASA

Economic Impact Study

2023

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UIC Nathalie P. Voorhees Center for Neighborhood and Community Improvement

Acknowledgments

Per request by PCI (Poarch Creek Indians) Productions, LLC, the Nathalie P. Voorhees Center for Neighborhood and Community Improvement (Voorhees Center) conducted economic impact analysis of the National Aeronautics and Space Administration (NASA), the agency's Moon to Mars (M2M) campaign, and investments in climate change research and technology for the Fiscal Year (FY) 2023.

The Voorhees Center is an applied research and technical assistance unit in the College of Urban Planning and Public Affairs at the University of Illinois Chicago. Its mission is to promote quality of life and wellbeing by assisting and working with local organizations, government entities and other stakeholders in efforts to understand and develop communities.

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We wish to thank everyone at NASA headquarters who contributed to the completion of this study. Our thanks to Gary Willis (Program Manager), Jason Shoemate, and Rebecca Patton for their effective management and coordination of this effort. We also thank Kristina LeGrand and colleagues for providing required data and for their clarifications to our many inquiries about technical data and information. We also thank others who provided feedback on the content and presentation of the report at various stages of the study.

Executive Summary

This study is an assessment of the economic impacts of the National Aeronautics and Space Administration (NASA), the agency's Moon to Mars (M2M) campaign, and investments in climate change research and technology for the Fiscal Year (FY) 2023. The assessment consists of three parts. The first part presents the estimation of NASA impacts on the U.S., each of the fifty (50) states, and Washington, D.C. The second and third parts analyze economic impacts attributable to the M2M campaign and the investments in climate change research and technology, respectively, on the same set of regions. The purpose of the economic impact assessment is to quantify the changes in employment, income, levels of business activity, and government revenue throughout the entire economy that result from NASA's activities, the M2M campaign and investments in climate change research and technology expenditures.

National Aeronautics and Space Administration (NASA)

NASA supports hundreds of thousands of jobs and contributes billions of dollars of tax base for states and the nation. The report estimates the economic benefits generated directly by NASA along with the very substantial additional economic activity created through the purchasing actions, labor income, and consumption spending that follow from the operations of the NASA centers and programs.

At the national level, NASA has 17,823 full-time equivalent (FTE) jobs, paying more than $3.5 billion in annual wages and benefits (Fiscal Year 2023).[1] Yet NASA's economic impact goes well beyond its immediate employment footprint. Wide-ranging benefits are created for local and state economies as well as the U.S. economy as a whole as NASA contracts for goods and services boost activity throughout the economy. The $23.3 billion in procurement (including P-card transactions) originating from NASA is very diverse, involving varied sets of manufacturing and service production sectors. Total NASA procurement and labor income amounts to $26.8 billion (including $3.516 billion of labor income, $23.218 billion of procurement activity, and $98 million of P-card spending).[2]

The impact estimates reported in this study are the sum of three channels of economic impact: (1) the direct contribution of NASA's own activities; (2) indirect (procurement) activities within NASA's U.S. supply chain; and (3) the induced effect that results as federal civil servants at NASA facilities, employees of contractors, and employees within the supply chain of those contractors spend their wages in the wider consumer economy. Some highlights of the economic impacts that occur throughout the national economy due to the activities of NASA include:

[1] The count of civil servants excluding those residing in U.S. territories is 17,821 FTEs. These are the figures used in the national economic impact model.

[2] NASA also had travel expenditure of $87.8 million that is not included in the modeling.

- The total employment sustained by NASA across the U.S. is estimated to be 304,803 jobs.[3]

- At the national level, NASA supports labor income of $27.6 billion per year and an estimated economic output of $75.6 billion annually. The approximately $90,547 in average annual labor income per job is 24% greater than the average across the U.S. economy ($73,416). The difference between average labor income (including wages and benefits) for a NASA civil servant ($197,283) and the U.S. average ($73,416) is much larger (269% of the U.S. average), reflecting the highly skilled and high-paid nature of NASA jobs compared to the average job in the U.S. economy.

- NASA generates an estimated $9.6 billion in annual federal, state, and local tax revenues throughout the U.S.

- For every FTE job located at a NASA facility, at least 16 additional jobs are supported throughout the U.S. economy.[4] For each million dollars of labor income earned by NASA civil service employees, an additional $6.8 million of labor income is generated in the U.S. And for each million dollars' worth of output produced by NASA, an additional $8 million of output—consisting of both intermediary inputs and consumption goods and services—is produced throughout the national economy. These figures are boosted by the large volume of procurement spending administered through NASA.[5]

- The top ten most impacted sectors account for 47% of total NASA employment impacts. Scientific research and development services is the most affected sector—19% of total NASA impacts are concentrated in this sector.

- NASA impacts are concentrated geographically as well. The top ten most impacted states account for 90% of total NASA employment impacts. These states are California, Texas, Florida, Alabama, Maryland, Virginia, Colorado, Ohio, Mississippi, and Washington. Impacts on Arizona, Massachusetts, Washington, D.C., Louisiana, and Utah are also relatively large. Figures 1 and 2 below illustrate the employment and output impacts of NASA across the states.

- Approximately 32% of overall NASA agency impacts in the U.S. are attributable to the M2M campaign. The M2M campaign-specific employment accounts for 3% of overall

[3] Jobs are reported as an annualized combination of full- and part-time positions based on the average output per employee for a given industry. That is part of the reason why we refer to impacts per year or annually occurring.

[4] "Supports" means partial or full support. We use "supported" instead of "created" to be conservative in our attribution of impacts to NASA. While NASA activities create new positions in some sectors of the economy, in many instances NASA helps sustain existing jobs in the economy.

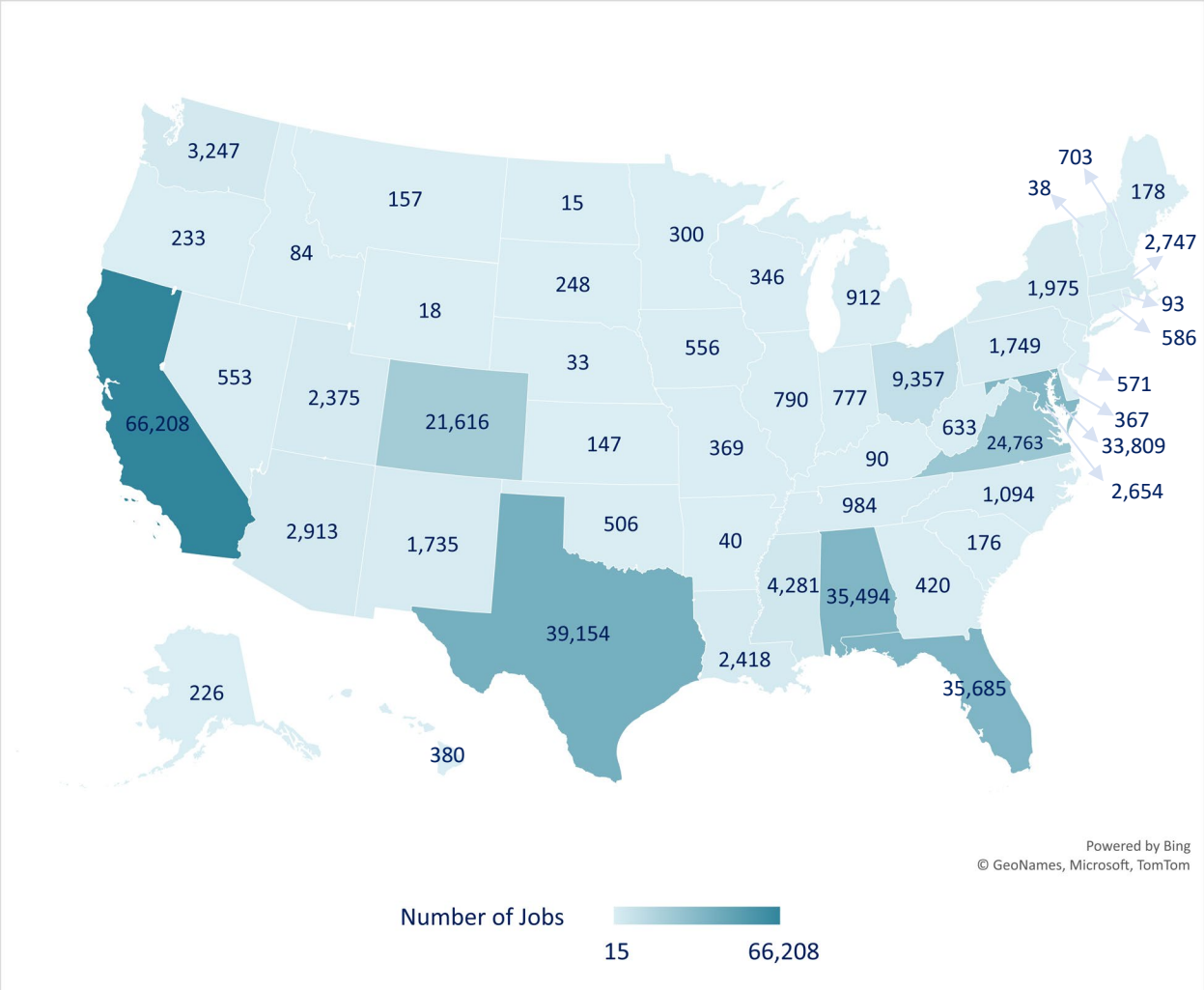
[5] The multipliers reported here are larger than is typical for this kind of study because of the volume of NASA procurement activity, all of which is classified as indirect impact. Additional multipliers calculated in the "conventional" way are reported in the Appendix. Two sets of multipliers are presented: one for NASA employment and one for NASA procurement spending.

NASA employment impacts while M2M campaign-related procurement accounts for 29% of overall NASA employment impacts.

- Approximately 11% of overall NASA agency impacts in the U.S. are attributable to the investments in climate change research and technology. Climate change research and technology specific employment accounts for 2% of overall NASA employment impacts while climate change research and technology-related procurement accounts for 9% of overall NASA employment impacts.[6]

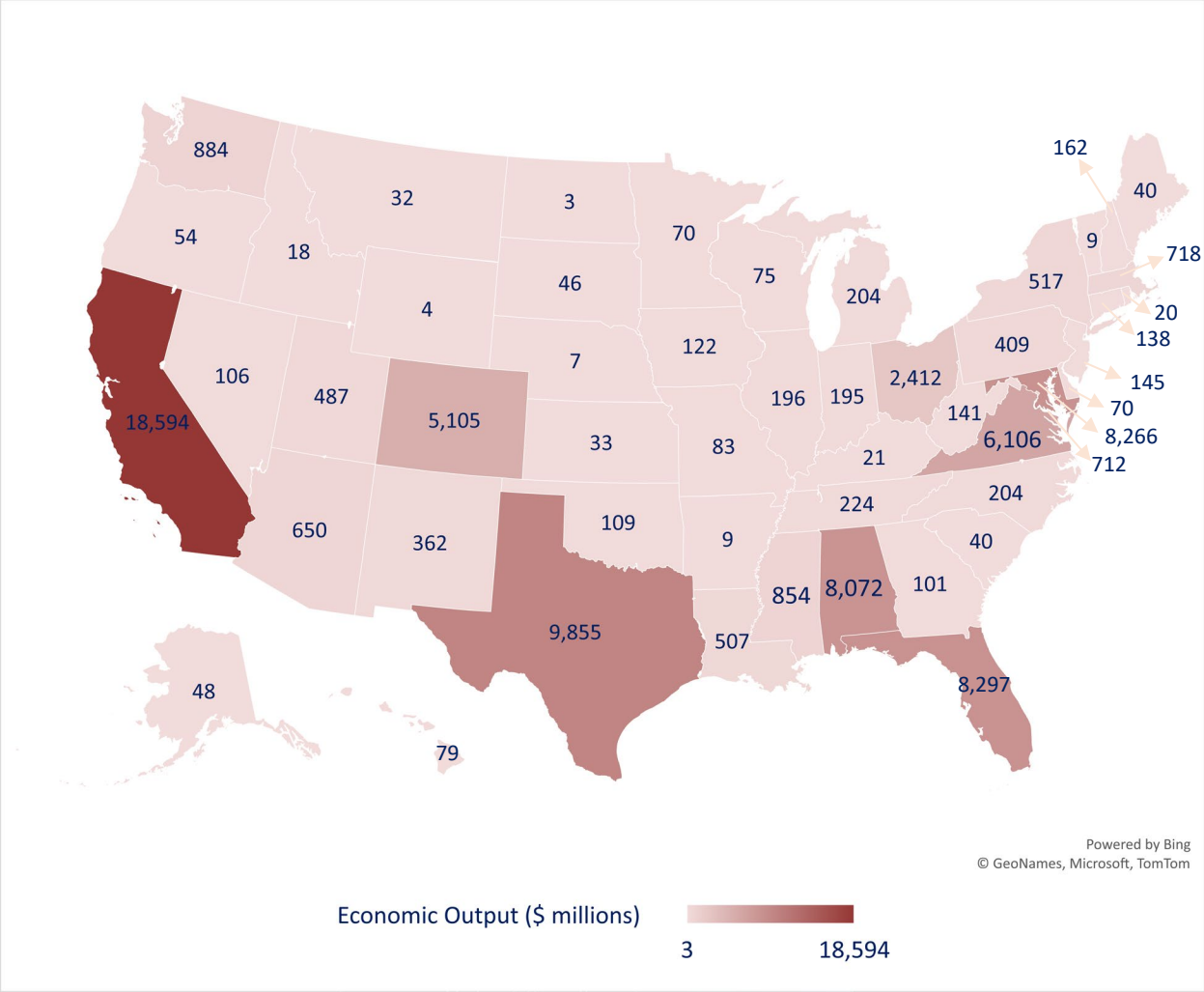
Figures 1 and 2 below illustrate the employment and output impacts of NASA across the states.

Figure 1: NASA Employment Impacts by State (employment consists of combination of full- and part-time positions)



[6] NASA impacts are inclusive of all M2M and Climate Change impacts discussed in the following pages.

Figure 2: NASA Output Impacts by State (in $ millions)



Moon to Mars (M2M) Campaign

NASA's human lunar exploration plans under Artemis include sending the first woman and first person of color to the surface of the Moon and establishing sustainable exploration by the end of the decade. Working with U.S. companies and international partners, NASA will make new scientific discoveries and lay the foundation for long-term lunar exploration and development. The agency will use what it learns on the Moon to prepare for humanity's next giant leap – sending astronauts to Mars.

It all starts with U.S companies delivering scientific instruments and technology demonstrations to the lunar surface, followed by a spaceship, called the Gateway, in orbit around the Moon that will support human and scientific missions, and human landers that will take astronauts to the surface of the Moon. The agency's powerful Space Launch System rocket and Orion spacecraft are the backbone to building the Gateway and transporting astronauts to and from Earth.[7]

[7] Information obtained from: https://www.nasa.gov/topics/moon-to-mars

NASA's Moon to Mars (M2M) campaign employs thousands of highly paid skilled professionals, and the U.S. government channels billions of dollars of federal spending on M2M into the states in the form of contracts. At the national level, NASA has 2,749 FTEs to support the M2M campaign, paying more than $712 million in annual wages and benefits. The M2M campaign's economic impact goes well beyond its immediate economic footprint represented by the employment of NASA civil servants. Wide-ranging benefits are created for the local and state economies as well as the U.S. economy as NASA's procurement for goods and services boosts economic activity elsewhere in the country. $7.7 billion in M2M procurement activity originating from NASA headquarters and centers involves almost every major category of manufacturing or service production. M2M contracts to firms, government agencies, and academic institutions not only support the U.S. aerospace industry but also result in advanced development and innovative solutions in areas including materials, structures, avionics, software, and analysis techniques.

Some highlights of the economic impacts that occur throughout the national economy from M2M activities include:

- The estimated amount of employment generated by M2M activities across the U.S. is 96,479 jobs.[8,9]

- The M2M campaign supports labor income of $8.6 billion per year and an estimated economic output of $23.8 billion annually.

- The M2M campaign generates an estimated $2.9 billion in federal, state, and local tax revenues throughout the U.S. each year.

- For every (FTE) civil service job located at NASA centers related to M2M, nearly 25 additional jobs are supported throughout the U.S. economy. For each million dollars of labor income earned by M2M-assigned NASA employees, an additional estimated $11.1 million of labor income is generated in the U.S. And for each $1 million worth of output produced by the M2M campaign, an additional $12.5 million worth of output—consisting of both intermediary inputs and consumption goods and services—is produced throughout the national economy. These figures are exceptionally large because of the extremely high volume of procurement spending relative to civil service employment involved in the M2M campaign.

- Figures 3 and 4 below illustrate the employment and output impacts of the M2M campaign across the states. Alabama, California, Texas, Colorado, Florida, Virginia, Washington, Utah, Maryland, and Ohio are the most impacted states, respectively. Ten states account for approximately 95% of all M2M-related employment impacts. As a share of overall NASA impacts, M2M campaign impacts are heavily concentrated in some states.

[8] The M2M impacts are completely a subset of the NASA impacts discussed in the previous page.
[9] Jobs are reported as an annualized combination of full- and part-time positions based on the average output per employee for a given industry.

Among the top-ten most impacted states (by the M2M campaign), more than half of overall NASA impacts in Washington, Utah, Colorado, and Alabama are attributable to the M2M campaign (Figure 5).

Figure 3: M2M Employment Impacts by State (employment consists of combination of full- and part-time positions)

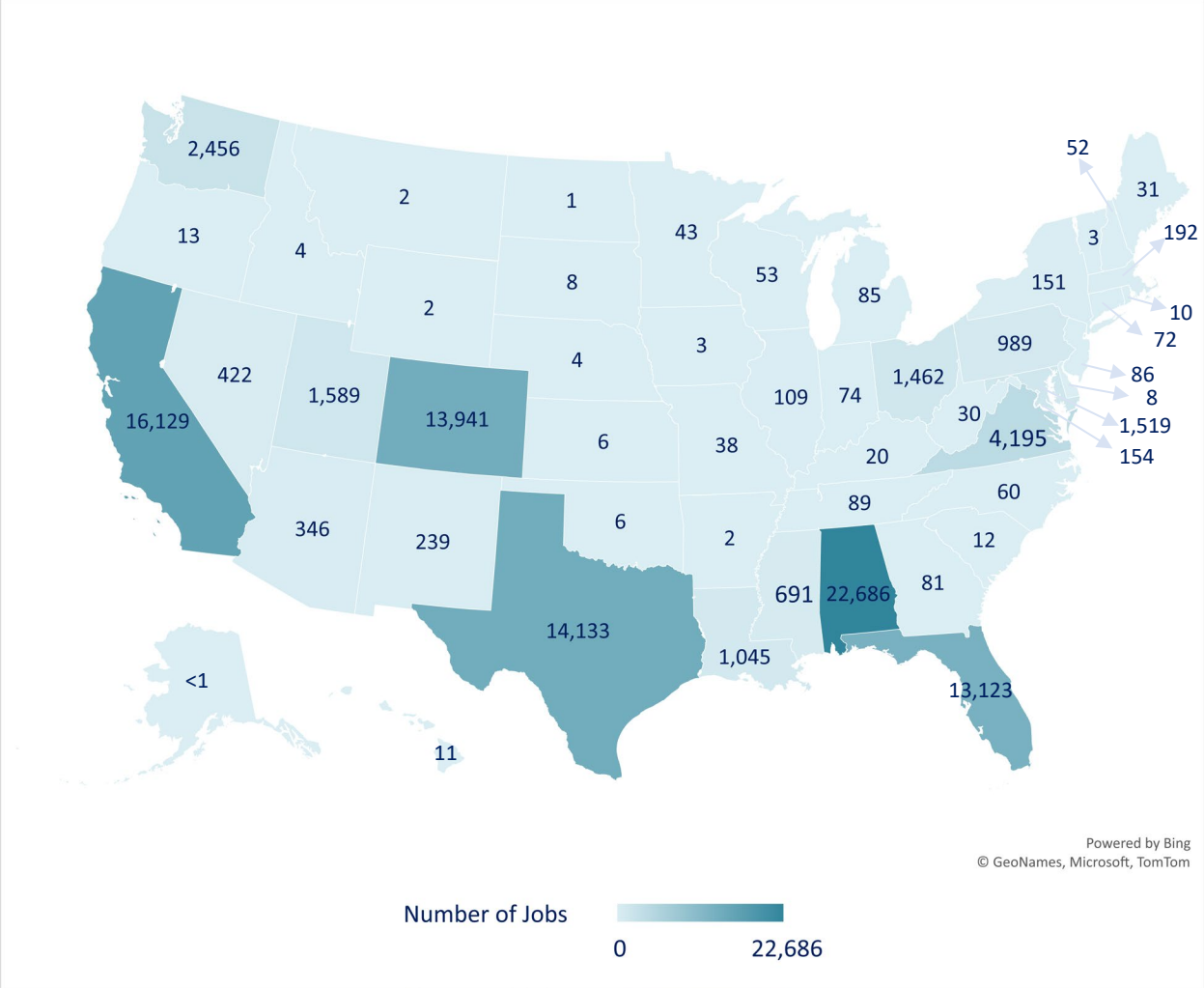


Figure 4: M2M Output Impacts by State (in $ millions)

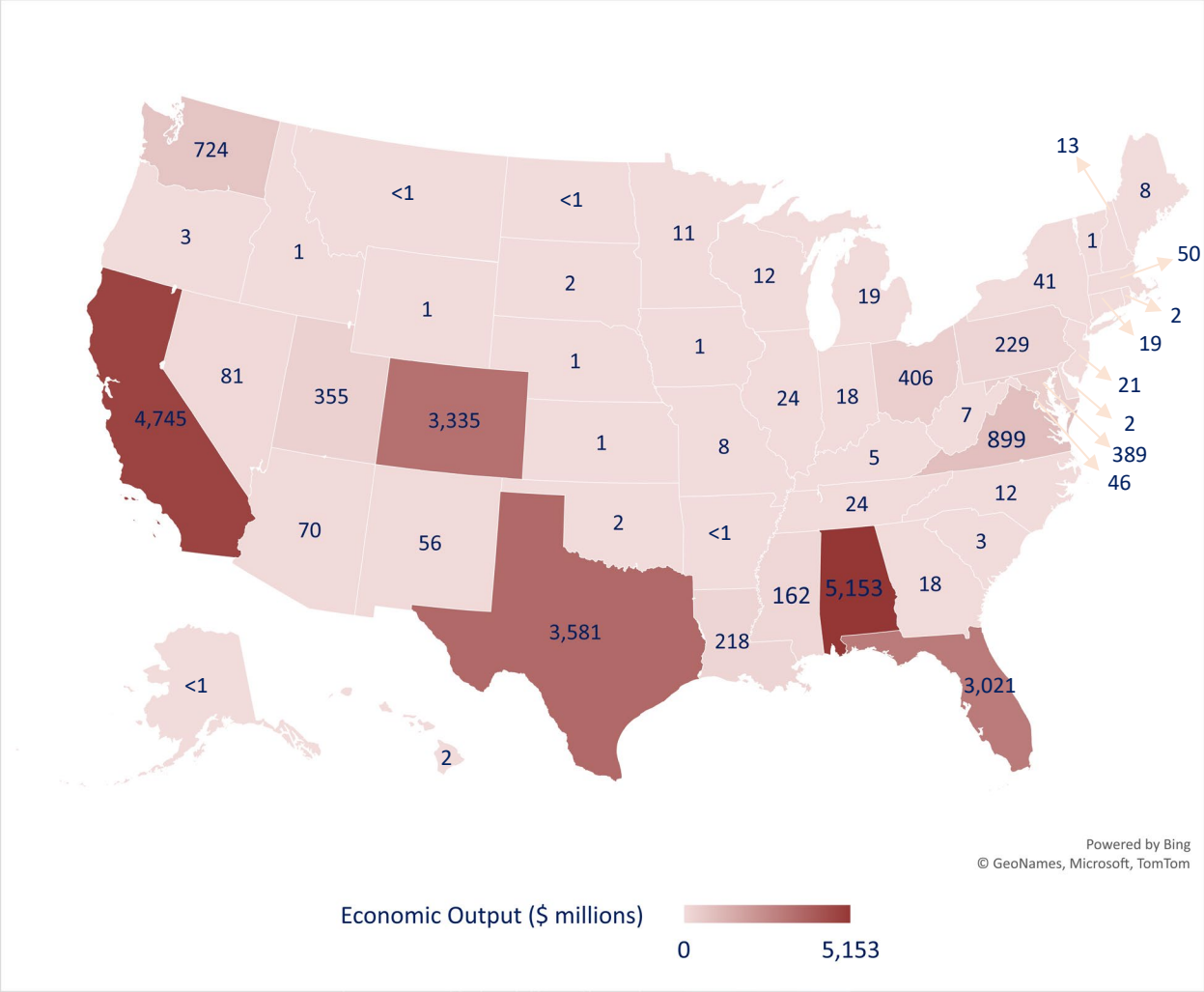
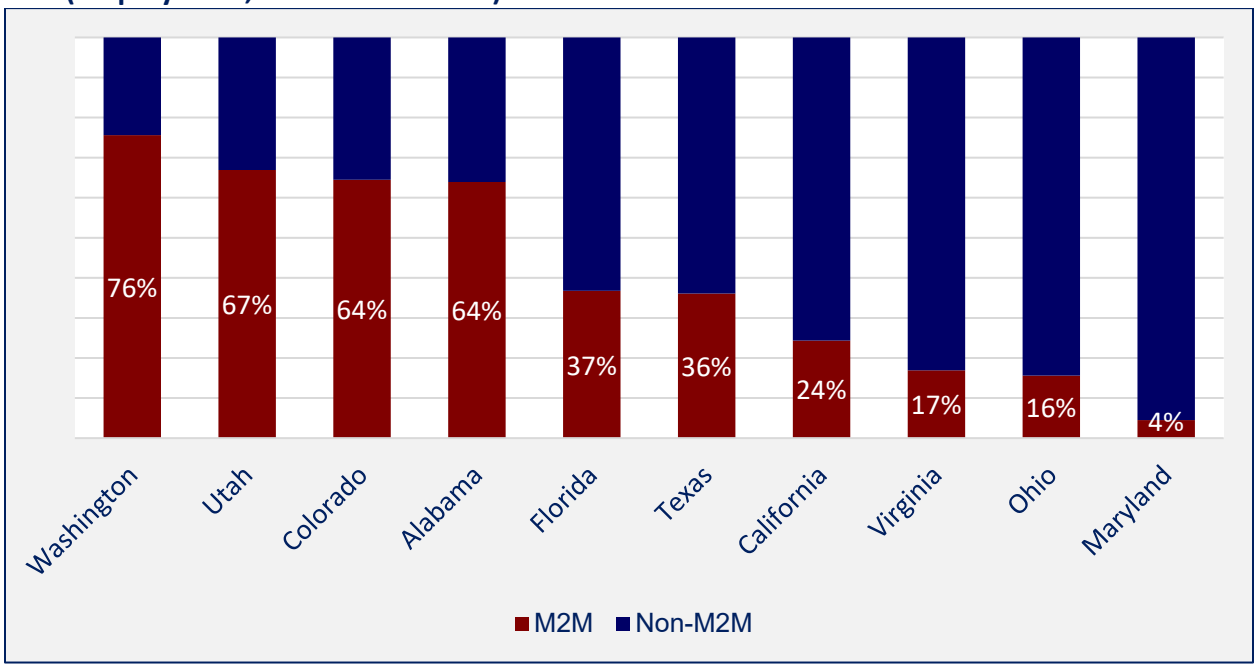


Figure 5: Share of M2M Campaign Impacts in Overall NASA Impacts for Top Ten States and U.S. (employment, based on FY 2023)



Investments in Climate Change Research and Technology

NASA is a global leader in studying Earth's changing climate. The agency's observations of Earth from space, the air, and on the ground provide information on how the interconnected systems of the planet interact.

NASA has a broad climate research program. Among the many areas NASA studies are greenhouse gases, temperature change, changes in sea ice and land ice, sea level rise, clouds and precipitation, and air pollution. NASA also develops technologies that can be used to mitigate or adapt to climate change, like sustainable aviation technologies.

In addition to providing the nation and world with unique climate observations, analysis, and modeling, this research helps NASA better assess the impacts of climate change on its mission and ensure the resiliency of its facilities and assets. NASA's capabilities in researching Earth and its atmosphere will continue to be critical in understanding causes and effects of temperature changes, sea level rise, and other major climate changes.[10]

NASA's investments in climate change research and technology employs thousands of highly paid skilled professionals, and the U.S. government channels billions of dollars of federal spending into the states in the form of contracts. At the national level, NASA has 2,009 FTEs to support the investments in climate change research and technology, paying more than $382 million in annual wages and benefits. The agency's investments in climate change research and technology go well beyond its immediate economic footprint represented by the employment of NASA civil servants.

[10] Information obtained from: https://climate.nasa.gov/nasa_science/history/

Wide-ranging benefits are created for local, state, and national economies as NASA's procurement for goods and services boosts economic activity elsewhere in the country. $2.4 billion in procurement originating from NASA headquarters and centers involve many different categories of manufacturing or service production. The agency's investments in climate change research and technology contracts to firms, government entities, and academic institutions help the U.S. maintain its global leadership role in collecting and disseminating systematic climate data that are increasingly critical to many aspects of life on Earth, from agriculture, to energy, and national security.

Some highlights of the economic impacts that occur throughout the national economy from NASA climate change research and technology investments include:

- The estimated amount of employment generated through NASA's investments in climate change research and technology across the U.S. is 32,900 jobs.[11, 12]

- The agency's investments in climate change research and technology support labor income of $2.9 billion per year and an estimated economic output of $7.9 billion annually.

- NASA's investments in climate change research and technology generate an estimated $1 billion in federal, state, and local tax revenues throughout the U.S. each year.

- For every (FTE) civil service job located at NASA centers related to investments in climate change research and technology, at least 15 additional jobs are supported throughout the U.S. economy. For each million dollars of labor income earned by investments in climate change research and technology-assigned NASA employees, an additional $6.7 million of labor income is generated in the U.S. And for each $1 million worth of output produced by NASA's investments in climate change research and technology, an additional $7.4 million of output—consisting of both intermediary inputs and consumption goods and services—is produced throughout the national economy.

- Figures 6 and 7 below illustrate the employment and output impacts of the agency's investments in climate change research and technology across the states. California, Maryland, Virginia, Colorado, Florida, Ohio, Alabama, Texas, Massachusetts, and New Mexico are the most impacted states, respectively. Ten states account for 81% of all climate change research and technology-related employment impacts. As a share of overall NASA impacts, investments in climate change research and technology impacts are heavily concentrated in some states. Among the top-ten most impacted states (by NASA investments in climate change research and technology), considerable portion of overall NASA impacts (10% or over) in New Mexico, Massachusetts, Ohio, Maryland,

[11] Climate change research and technology impacts are completely a subset of the NASA impacts discussed earlier.
[12] Jobs are reported as an annualized combination of full- and part-time positions based on the average output per employee for a given industry.

Virginia, Colorado, and California are attributable to the investments in climate change research and technology (Figure 8).

Figure 6: NASA's Investments in Climate Change Research and Technology Employment Impacts by State (employment consists of combination of full- and part-time positions)

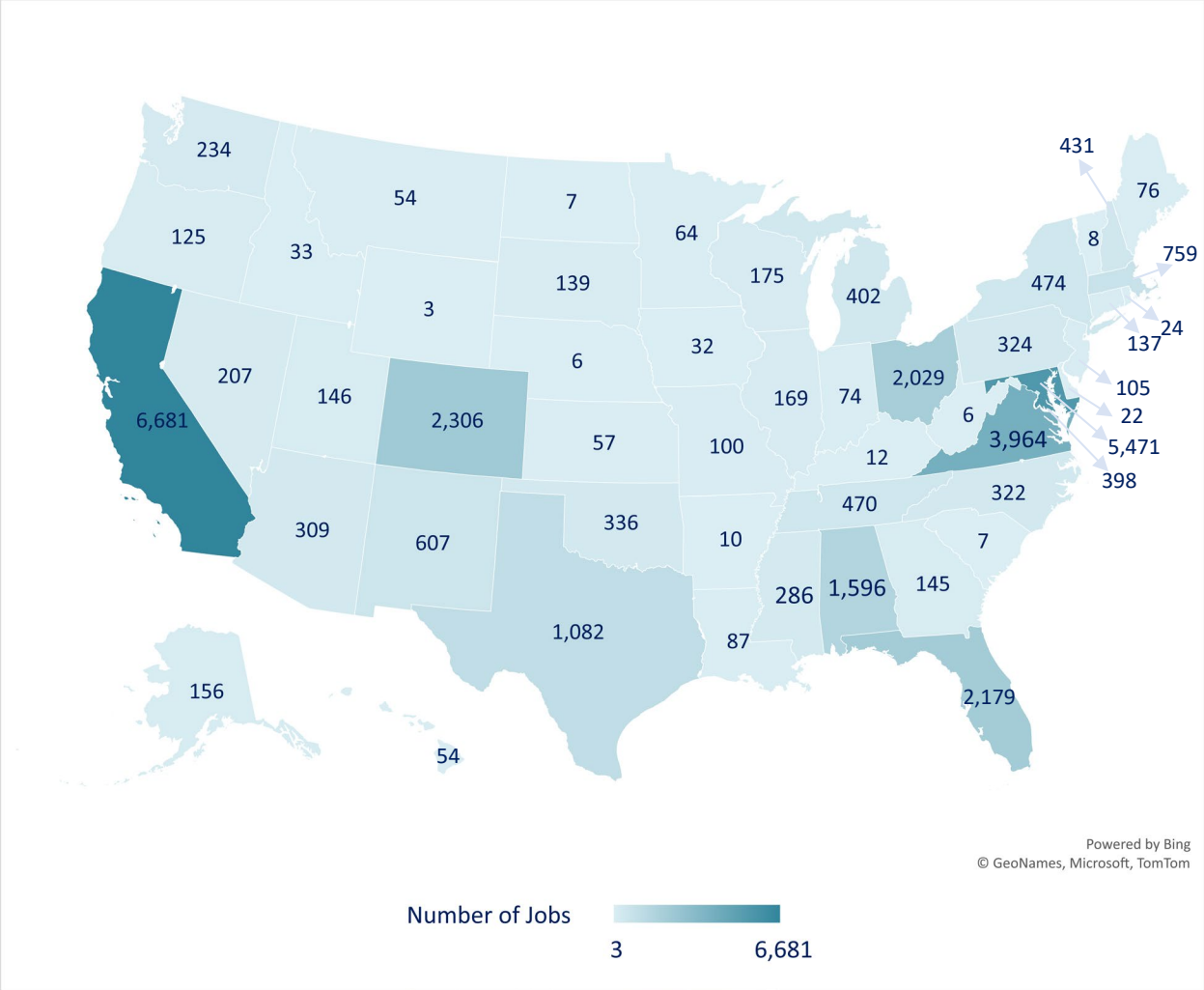


Figure 7: NASA's Investments in Climate Change Research and Technology Output Impacts by State (in $ millions)

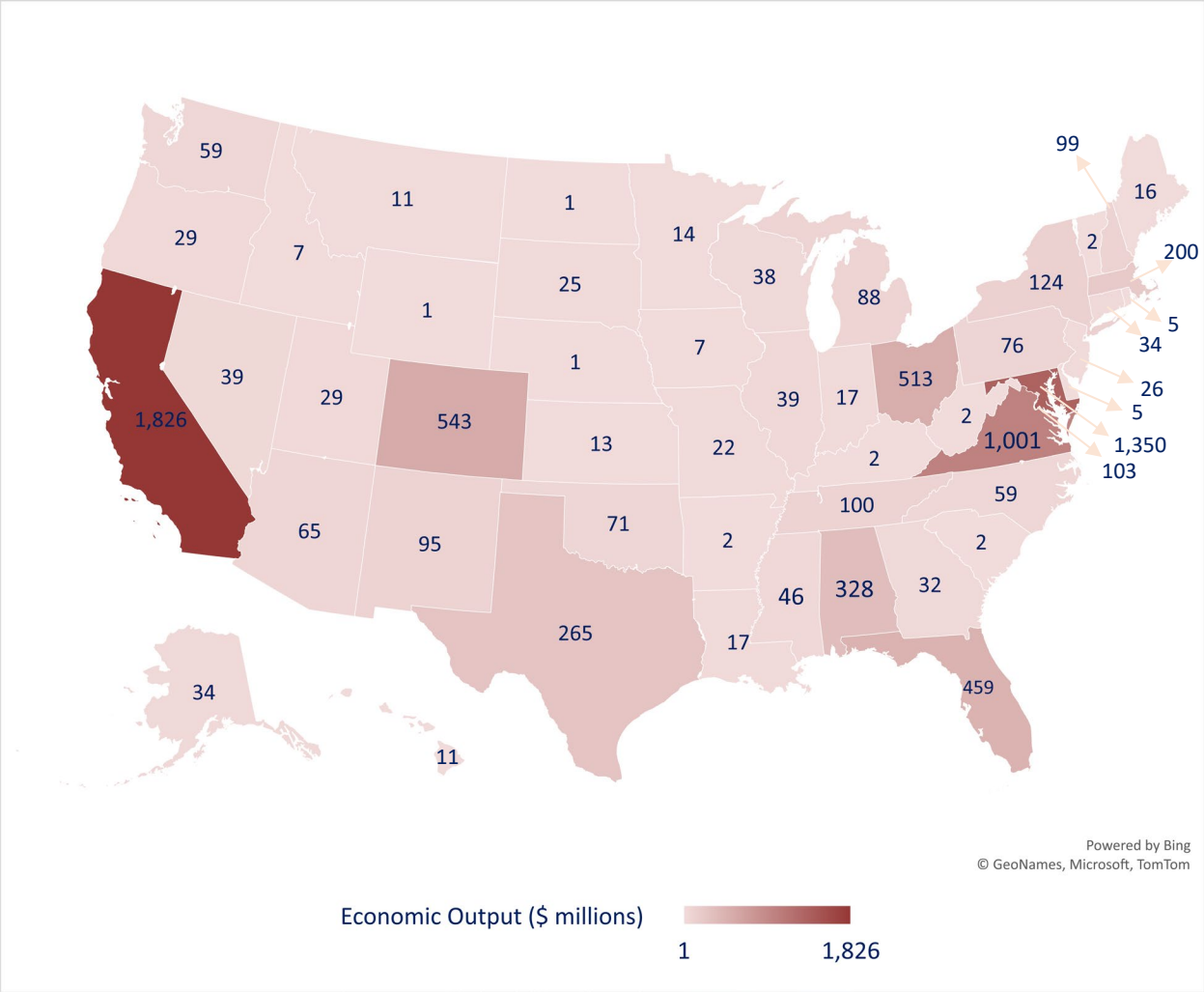
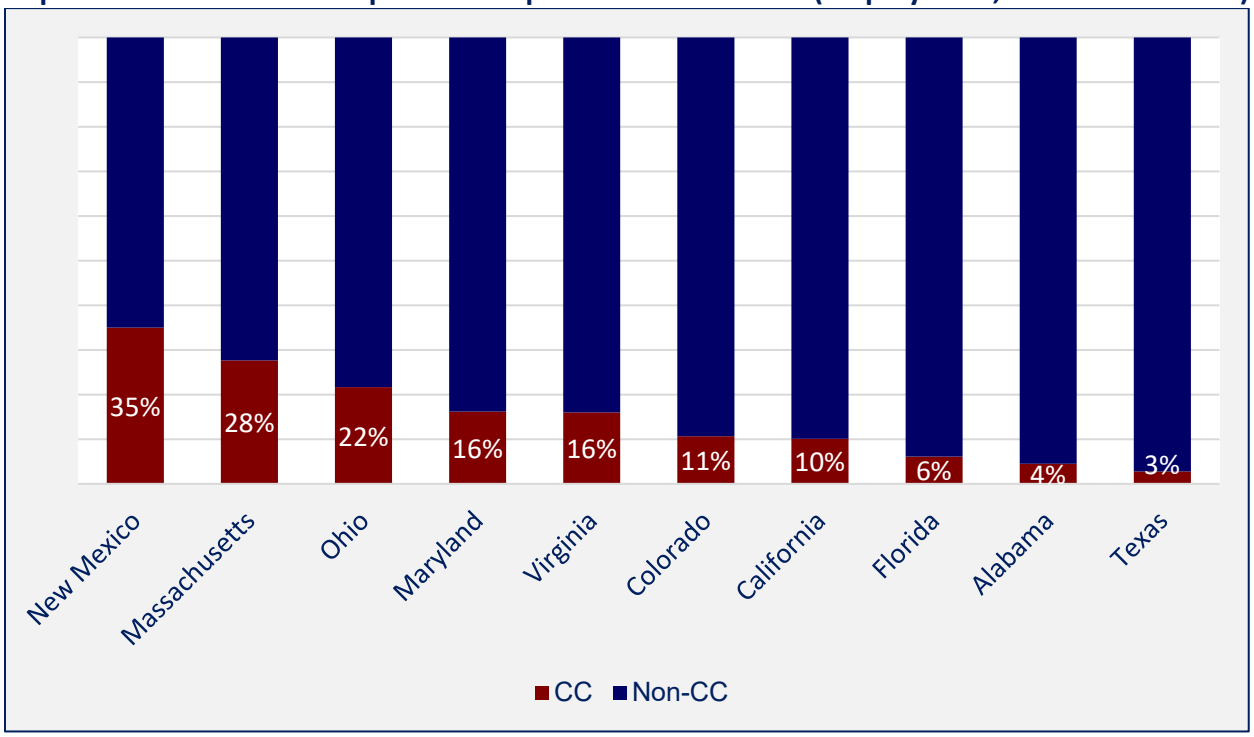


Figure 8: The Share of NASA's Investments in Climate Change Research and Technology Impacts in Overall NASA Impacts for Top Ten States and U.S. (employment, based on FY 2023)



INTRODUCTION

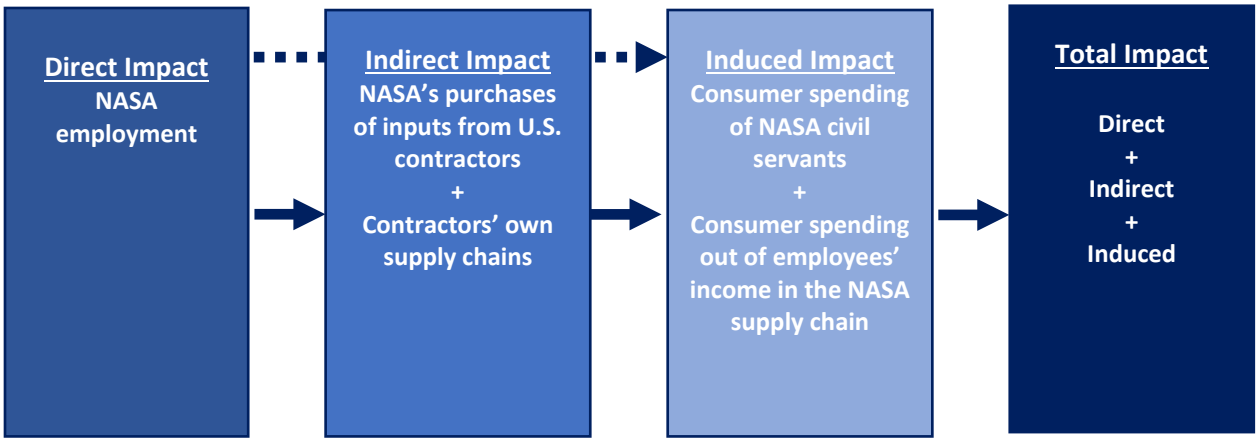
This study is an assessment of the economic impacts of the National Aeronautics and Space Administration (NASA), the Moon to Mars (M2M) campaign, and the investments in climate change research and technology for the Fiscal Year 2023. The assessment consists of three parts. The first part presents the estimation of NASA impacts on the U.S. as a whole, each of the fifty (50) states, and Washington, D.C. The second part analyzes economic impacts attributable to the M2M campaign in the same set of regions. The third part analyzes economic impacts attributable to the investments in climate change research and technology in the same set of regions. The purpose of the economic impact assessment is to quantify the changes in employment, income, levels of business activity, and government revenue throughout the economy that result from NASA's activities, the M2M campaign, and from investments in climate change research and technology.

We present direct, indirect, and induced employment impacts (see Box 1) along with output, labor income, and tax revenue impacts for each of the geographic regions (see Box 2).

Box 1: Types of Economic Impact

- ***Direct impacts*** **–** the activity or event to be assessed, typically specified as a change in demand for the output of an industry or a change in an industry's employment.
- ***Indirect impacts*** **–** economic activity resulting from the increase or decrease in input purchases within the region by the industry directly impacted in response to the change in demand or employment. In the context of this study, indirect effects are the purchases of goods and services by government agencies and private sector contractors as well as by the industries that supply them. Examples include an aerospace firm carrying out contract work for NASA and that aerospace firm's purchases of electronic components and engineering consulting services from other firms to enable its own production.
- ***Induced impacts*** **–** economic activity resulting from all affected industries paying wages and benefits to their employees and allocating profit to their owners, some portion of which is spent within the region on consumer goods and services.
- ***Total impacts*** **–** the sum of direct, indirect, and induced impacts (see Figure 1).

Figure 1: Direct, Indirect, Induced, and Total Economic Impacts



Note: *This figure is adapted from Oxford Economics (2015).*

Box 2: Types of Economic Activity

- ***Employment –*** the annual average of monthly jobs in the economy/industry. The number consists of both *full-time and part-time* jobs.
- ***Labor Income –*** total of all forms of employment income for a year, including employee compensation (wages and benefits) and proprietor income.
- ***Value Added –*** the difference between an industry's, or an establishment's, total output, and the cost of its intermediate inputs for a year. Value added is, equivalently, payment to labor (employee compensation) and rent to capital (profit) plus business taxes. Value added is a measure of the contribution to gross domestic product (at the national, state, or local scale) made by an individual producer, industry, or sector. Direct value added is calculated using the estimated ratios of value added-to-employee in the following industries: (i) guided missile and space vehicle manufacturing, (ii) propulsion units and parts for space vehicles and guided missiles.
- ***Output –*** the value of annual industry production. Output is reported in producer prices, separating margins such as for wholesale distribution, transportation, and retail into those respective industries. Production at NASA facilities has three components: (1) The direct output component represents the value of all activities at NASA (research and development, production, and management), including both those outputs that are sold to consumers (final consumption) and those that are inputs into further production (intermediary products and services). Direct output is calculated using the estimated ratios of output-to-employee in the following industries: (i) guided missile and space vehicle manufacturing, (ii) propulsion units and parts for space vehicles and guided missiles (2) The indirect component is the value of the goods and services produced in industries that supply the activities at NASA facilities, and the value of the goods and services that are in the supply chain of those industries. (3) The induced component is the value of goods and services, both intermediary and final consumption, that are produced in response to labor income spending of employees in all affected industries. Please note that the term production as used here is not equivalent to product as used Gross Domestic Product (GDP) for Value Added (see above).
- ***Tax Revenues –*** the annual taxes collected by federal, state, and local governments through income, payroll, import, corporate, and property taxes, as well as some forms of licenses. Tax revenue estimates are generated from the application of average (per capita and per employee) tax rates and license fees averaged across multiple jurisdictions within a study region and should be treated as only rough approximations.

The modeling approach adopted in this study is designed to represent outcomes of complex economic relations in aggregate form to enable ready connections between impact-generating event categories and their impacts. Accordingly, beside estimates of the total economic impacts, the report provides impact multiplier factors (or multipliers for short) for different types of outcomes (see Box 3) that can be used to quickly assess the total scope and sectoral impact of relatively small employment and procurement changes in the future. The shift in the triggering or direct event—such as an increase in employment or a decrease in procurement amount—can simply be multiplied by the relevant outcome multiplier to produce an estimate of the total effect magnitude (direct, indirect, and induced). This procedure is appropriate for situations in which major economic features and relationships, such as employee location patterns and industry purchasing linkages, are not expected to change. Small shifts are generally tolerated, but large changes in the economy may alter inter-industry relationships and require model reevaluation. (The threshold where changes shift from small to large is determined by analyst judgment.)

Box 3: Multipliers

- ***Employment multiplier*** – the estimated ratio of the change in total local employment to the change in direct employment. For example, an employment multiplier of 3.00 indicates that the creation of 1 direct new job is expected to support 2 additional jobs in the local economy, for a total impact of 3 new jobs.
- ***Income multiplier*** – the estimated ratio of the change in total income throughout the local economy to the change in income from direct employment. It is calculated analogously to the employment multiplier. For example, an income multiplier of 2.75 indicates that a $1.00 change in income in the industry directly affected by the events being considered is expected to result in an additional income of $1.75 throughout the local economy, for a total income impact of $2.75.
- ***Value added multiplier*** – the estimated ratio of the change in total value added throughout the local economy to the change in value added from direct employment. The value-added multiplier is calculated analogously to the employment, income, and output multipliers.
- ***Output multiplier –*** the estimated ratio of the change in total output (i.e., gross sales and additions to inventory) throughout the economy to the change in output from direct employment. The output multiplier is calculated analogously to the employment, income, and value added multipliers.

An input-output modeling framework is used to estimate the impacts. We utilize IMPLAN (IMpact analysis for PLANning) software and 2022 industry data, the most current dataset available at the initiation of the study (see Box 4).

Box 4: IMPLAN Industry Data

We used 2022 industry data (economic modeling data from IMPLAN), FY2023 NASA procurement data (provided by NASA), and FY2023 NASA labor income data (provided by NASA). The economic modeling data are the national and regional estimates of industry-to-industry purchasing and spending interactions that are produced by IMPLAN.

If we had used 2023 rather than 2022 industry data, the results of the impact analysis would be unlikely to be significantly different. IMPLAN model data updates are relatively minor except for every five years when the company makes substantial changes to its database based on the benchmark input-output (I-O) accounts developed and released by the Bureau of Economic Analysis (BEA). (Even then, most industry supply and purchasing relationships are quite consistent over time, only a few industries change substantially more rapidly.) The 2022 data we used in this study incorporate the latest updated industry statistics and benchmark I-O accounts of BEA. Therefore using 2023 data would not impact the estimates provided in this study significantly.

The benchmark I-O accounts are one of the major elements of the U.S. national and industry economic accounts. They provide detailed statistics on economic processes and relationships, and they provide essential information for other economic accounts. They are used to set the level of Gross Domestic Product (GDP) in the NIPAs (national income and product accounts), and they provide commodity detail on the composition of the final-use categories. In addition, they provide information on which inputs industries use to produce their output and which commodities are produced by each industry. The benchmark I-O accounts consist of make tables, use tables, and direct and total requirements tables. They are prepared at 5-year intervals, primarily based on Economic Census data.

pre-pandemic (i.e., 2019). Specifically, labor productivity in some industries is substantially greater in the 2022 economy. This increased productivity leads to reduced employment impacts in some industry sectors that are key to the analysis and also to smaller estimated employment multipliers. As a result, jobs estimates resulting from IMPLAN 2022 will likely be less than from previous versions of IMPLAN given the same inputs. The productivity growth was most likely driven by two actions. First, federal investments made through the Biden Administration's Build Back Better Framework and related programs boosted productivity across various sectors of the economy. Second, many companies eliminated redundant and low-performing positions to improve efficiency between 2019 and 2022.

National Aeronautics and Space Administration (NASA)[13]

The National Aeronautics and Space Administration (NASA) is responsible for unique scientific and technological achievements in human spaceflight, aeronautics, space science, and space applications that have had widespread national and global impacts. Established in 1958, NASA has been the center of U.S. civil aerospace research and development.

In addition to agency leadership (Headquarters), NASA has research and flight centers and facilities with unique capabilities:

[13] Information is obtained from https://www.nasa.gov/content/nasa-history-overview and https://www.nasa.gov/about/org_index.html.

NASA Research and Flight Centers

- Ames Research Center (IT, fundamental aeronautics, bio, and space science technologies)
- Armstrong Flight Research Center (Flight research)
- Glenn Research Center (Aeropropulsion and communications technologies)
- Goddard Space Flight Center (Earth, the solar system, universe observations, and space communications and navigation)
- Johnson Space Center (Human space exploration)
- Kennedy Space Center (Prepare and launch missions around the Earth and beyond)
- Langley Research Center (Aviation, space technology, and Earth science)
- Marshall Space Flight Center (Space transportation and propulsion technologies)
- Stennis Space Center (Rocket propulsion testing and remote sensing technology)
- Goddard Institute for Space Studies (Broad study of global climate change)

NASA Facilities and Labs

- Jet Propulsion Laboratory (Robotic exploration of the solar system)
- Katherine Johnson Independent Verification and Validation Facility (Provides safety and cost-effectiveness for mission critical software)
- Michoud Assembly Facility (Manufacture and assembly of critical hardware for exploration vehicles)
- NASA Engineering and Safety Center (Independent testing, analysis, and assessments of NASA's high-risk projects)
- NASA Safety Center (Development of personnel, processes and tools needed for the safe and successful achievement of strategic goals)
- NASA Shared Services Center (Financial management, human resources, information technology, and procurement)
- Wallops Flight Facility (Suborbital Research Programs)
- NASA Goddard Institute for Space Studies (Broad study of global climate change)
- NASA White Sands Test Facility (Engine and hazardous materials, components and systems testing and analysis)

Results from NASA's research and development activities ripple throughout the national economy, supporting high-tech industries and creating or sustaining tens of thousands of knowledge-intensive jobs. In addition, NASA invests in economically valuable technologies that help the nation maintain its competitive advantage. NASA develops hundreds of new technologies each year and transfers thousands of products, services, and processes to private businesses. These transfers improve U.S. businesses productivity and global competitiveness. Given the multidimensional nature of the benefits originating from NASA facilities, no single metric can capture the returns from its activities. This report focuses on NASA's economic impacts that can be reliably and precisely quantified through economic input-output models.

Moon to Mars (M2M) Campaign[14]

NASA's human lunar exploration plans under Artemis call for sending the first woman and first person of color to the surface of the Moon and establishing sustainable exploration by the end of the decade. Working with U.S. companies and international partners, NASA will uncover new scientific discoveries and lay the foundation for long-term lunar exploration and development. The agency will use what it learns on the Moon to prepare for humanity's next giant leap – sending astronauts to Mars.

It all starts with U.S companies delivering scientific instruments and technology demonstrations to the lunar surface, followed by a spaceship, called the Gateway, in orbit around the Moon that will support human and scientific missions, and human landers that will take astronauts to the surface of the Moon. The agency's powerful Space Launch System rocket and Orion spacecraft are the backbone to build the Gateway and transport astronauts to and from Earth. The following represents the variety of programs and projects of the M2M campaign:

- The Space Launch System
- Orion spacecraft
- All ground support equipment for development, testing, and launch of SLS/Orion
- Gateway
- Human Landing System
- Space suits for lunar surface operations
- In situ resource utilization, surface power systems, life support, and other advanced technology developments for a sustained lunar presence
- Mars 2020 & Sample Return
- Commercial Lunar Payload Services

For additional information about NASA's Moon to Mars campaign, please visit EXPLORE MOON to MARS.[15]

Investments in Climate Change Research and Technology[16]

NASA is a global leader in studying Earth's changing climate. The agency's observations of Earth from space, the air, and on the ground are helping humans learn how the interconnected systems of the planet interact.

NASA has a broad climate research program. Among the many areas NASA studies are greenhouse gases, temperature change, changes in sea ice and land ice, sea level rise, clouds and

[14] Information is obtained from https://www.nasa.gov/topics/moon-to-mars
[15] The EXPLORE MOON to MARS website address is https://www.nasa.gov/topics/moon-to-mars.
[16] Information is obtained from https://climate.nasa.gov/nasa_science/history/

precipitation, and air pollution. NASA also develops technologies that can be used to mitigate or adapt to climate change, like sustainable aviation technologies.

In addition to providing the nation and world with unique climate observations, analysis, and modeling, this research helps NASA better assess the impacts of climate change on its mission and ensure the resiliency of its facilities and assets. NASA's capabilities in researching Earth and its atmosphere will continue to be critical in understanding causes and effects of temperature changes, sea level rise, and other major climate changes. The following represents the variety of programs and projects included in NASA's investments in climate change research and technology:

- Earth Science Activities
  - *Earth Science Research*
  - *Earth Systematic Mission*
  - *Earth System Science Pathfinder Applied Science (Pathways, including SERVIR)*
  - *Earth System Explorers*
  - *Earth Science Technology*
- Aeronautics Activities
  - *Advanced Air Transport Tech*
  - *Advanced Composites Project*
  - *Advanced Air Mobility*
  - *Air Traffic Management Exploration*
  - *Airspace Technology Demonstration*
  - *Convergent Aeronautics Solutions*
  - *Cross Program Operations*
  - *Environmentally Responsible Aviation*
  - *Hi-Rate Composite Aircraft Manufacturing*
  - *Hybrid Thermally Efficient Core*
  - *SMART-NAS Test Bed for Safe TBO*
  - *Transformational Tools and Technologies*
- Space Technology Activities
  - *Portions of Space Technology Research Grants; Technology Transfer; Small Business Technology Transfer*
  - *TDM Fission Surface Power*
  - *GCD Nuclear Systems*
- Portions of EPSCoR
- Construction and environmental compliance and remediation efforts at NASA facilities

For more information about NASA's investments in climate change research and technology, please visit GLOBAL CLIMATE CHANGE, Vital Signs of the Planet.[17]

[17] The GLOBAL CLIMATE CHANGE website address is https://climate.nasa.gov.

Report Organization

The extensive activities of NASA and the unique capabilities of the Moon-to-Mars campaign and the investments in climate change research and technology encompass significant research and development (R&D), as well as manufacturing in aerospace and related industries that require the hiring of thousands of highly skilled scientists and engineers. NASA expends millions of dollars in many states in the form of direct employment and contracts with private sector vendors. This infusion of federal spending in the form of labor income and procurement in turn supports thousands of additional jobs and millions of dollars of state and local government tax revenues across the country. This study quantifies these economic impacts and analyzes their significance for the economies involved.

The report is organized as follows. Immediately following this introduction, Section 2 surveys the pertinent literature and provides a brief theoretical background for the modeling process, describing the steps taken to conduct the economic impact analysis. Section 3 describes the data used in the study, examining NASA employment at various research and flight centers (including information related to labor income and employee residences) along with contract awards originating from NASA and those related to the M2M campaign and the investments in climate change research and technology (including the geographic distribution of these contracts). Understanding the data is key to understanding the approach and for interpreting the results of the economic impact analysis. Section 4 discusses the modeling approach, employing data and software from IMPLAN. Section 5 examines additional issues surrounding the impact estimates, focusing largely on the geographic designation of impact regions and the specification of economic activities and events—the inputs into the models. These two sections (4 and 5) detail the modifications we made to the basic IMPLAN model and the most important assumptions that underlie the impact estimates, as well as the rationales behind them. Finally, Section 6 provides estimates of the economic impacts generated by NASA, the M2M campaign, the investments in climate change research and technology, and analyzes the meaning of these impacts for the economies involved.

LITERATURE SURVEY

The Conceptual Bases of Economic Impact Analysis

Economic impact analysis involves applying a change in final demand to a descriptive economic model and then analyzing the resulting changes in the economy. A concise definition of impact analysis is that it is an assessment of the overall changes in an economy that occur as a result of one or more changes in specific economic activities (IMPLAN, 2004).

Export base theory (also termed economic base theory) provides the conceptual foundation for input-output economic impact assessment models. The first of two fundamental concepts that undergird export base theory is that an area's economy can be divided into two types of economic units: (1) basic and (2) non-basic. The ***basic sector*** is defined as those firms that sell goods and services to markets outside the local area. The revenue received by basic sector firms for their exports of goods and services is termed basic income. The remainder of the area's economy consists of those firms that supply goods and services to customers within the area. These firms are referred to as the ***non-basic sector*** or sometimes as ***residentiary*** or ***local trade and service*** activities.

The second key concept of export base theory is that the level of non-basic activity in an area is determined by the level of basic activity, and a given change in the level of basic activity brings about a predictable change in the level of non-basic activities. This relationship is known as the ***multiplier*** effect. Thus, export base theory emphasizes external demand for the products of the basic sector as the principal force determining change in an area's level of economic activity (Leistritz, 1994).

The basis for the multiplier effect is the interdependence of (or linkages between) the basic and non-basic sectors within an area's economy. As the basic sector expands, it requires more inputs (for example, labor and supplies). Some of these inputs are purchased from local firms and households. As the firms in the non-basic sector expand their sales to the basic sector, they too must purchase more inputs, their suppliers in turn must increase their own input purchases, and so on. Increased wages and salaries paid to labor and management in the basic sector, together with similar increases in the non-basic sector, lead to increases in the incomes of area households. Some of this additional income is spent locally for goods and services, some is saved, and some leaves the area economy as payments for imported goods and services (or as additional tax payments to government). To the extent that additional income is spent locally for goods and services, the output of local firms increases further, resulting in additional cycles of input purchases, income increases and consumption expenditures. This cycle of spending and re-spending within the local economy is the basis of the multiplier effect (Leistritz & Murdock, 1981). The magnitude of the multiplier effect is determined by the proportions of a given dollar of additional input purchases and a given dollar of income spent locally. High multiplier values are associated with high levels of local purchases and spending, which typically reflect a diversified, relatively self-sufficient economy. Larger regions tend to have higher multiplier values.

The Principles of Economic Impact Analysis

To begin the process of estimating economic impacts, it is important to identify the counterfactual scenario against which the specific changes in economic activity are contrasted (Drucker, 2015). For example, if local industries are expected to increase their activity due to a rise in demand for their products or services, the source of that demand—the economic agent or agents seeking the additional products or services—could be considered to have made a choice versus seeking a different set of products or services or obtaining them instead from industries in a different location. In this study, wherein the federal employees' labor income at NASA facilities and the procurement originating from NASA constitute the changes in economic activity that are examined, we consider these changes relative to a hypothetical situation of zero NASA employment and procurement activity. Although this counterfactual of no activity is unrealistic on a nationwide basis in that the federal government resources currently applied to NASA would likely be expended for other tasks or through different federal agencies, a hypothetical baseline scenario of zero replacement activity is a common strategy for impact analyses seeking to obtain measures of the entire economic impact of a particular institution and its associated activities. For smaller geographic regions, such as a state, the counterfactual scenario of zero activity is more realistic because there is no reason to suppose that the allocation of federal resources in the absence of the NASA would concentrate in that specific state. This principle applies to the M2M campaign and the investments in climate change research and technology as well.

After determining the counterfactual scenario, the next step is to identify the topic or topics of concern as a set of economic events defined by expenditures. These expenditures constitute the initial changes that stimulate further activity throughout the economy. The actions and the economic activity they stimulate together comprise the impacts. Although impact modeling software (such as IMPLAN) provides a framework to conduct an analysis of economic impacts, each stage of an analysis should be carefully scrutinized to make sure it is logical. Procedures and assumptions ought to be validated prior to entry into the model (IMPLAN, 2004). Furthermore, understanding how an industry and the various parts of its supply chain respond to changes in demand and judging the validity of the assumptions made as part of operationalizing a regional input–output model are essential to usefully interpreting the outputs of an economic impact model (Drucker, 2015; Low & Isserman, 2009). The key assumptions and their implications are detailed in the following sections of the report.

Economic impact analysis is a projection process, and the figures generated should be regarded as reasonable approximations and estimates rather than as accurate predictions. There are several points in the conduct of an economic impact analysis where modeling choices and underlying assumptions substantially affect the final results. To avoid inadvertent errors or misapplications, economic impact analyses should include clear documentation of the procedures applied to obtain the initial events modeled, key underlying assumptions, and the rationales for both (Crompton *et al.,* 2001).

The Importance of Economic Impact Analysis to Public Policy Making

Economic impact assessment has developed as a pragmatic approach that attempts to bring appropriate concepts and tools from regional economics and public finance to bear on problems of concern to policy makers (Leistritz, 1994). Practitioners use impact analysis for different purposes. For example, it might be used to measure the impact of a new factory moving into an area. Governments use impact analysis to inform and guide policy decisions and planning. Researchers use impact analysis to study relationships among different elements in an economy (IMPLAN, 2004).

The increasing demand for economic impact analyses by public agencies can be explained in two ways. First, economic impact assessments are sought by policy makers and resource managers because they address issues that are key to a wide variety of decisions. For example, in determining whether to designate certain public lands as wilderness areas, land managers may need to consider the economic and fiscal impacts of alternative land uses (e.g., wilderness versus ranching or mining). When large-scale mining and resource development projects have been proposed, the local economic and fiscal impacts often have been one of the principal topics of debate.

Second, as state and local governments become more heavily involved in economic development efforts, economic impact analysis tools can be useful in establishing priorities for incentive programs. While a number of states are now using selected measures of direct economic impact (most commonly the number of jobs created) as criteria in awarding financial support, the total economic impact (including secondary effects) is probably a more meaningful criterion (Leistritz, 1994). This also explains why a key purpose of economic impact studies is to measure the economic return to residents (Crompton *et al.,* 2001). Public interest in economic impacts is ultimately a concern about people and their standard of living (Siegfried *et al.,* 2007).

DATA ANALYSIS

Description of NASA Data

NASA supplied information about activities at NASA headquarters and research and flight centers including salary, benefits, and total labor income in all 50 states as well as in Washington, D.C. The data contain places of residence for NASA employees living in different states (with all individual identifiers removed). Procurement data for NASA list contract amounts awarded to vendors located in different parts of the country with a brief description of activities or services that vendors were awarded to carry. NASA provides funding not only to private sector firms, but also to colleges and universities, federal government agencies, and civic organizations in the form of research and development contracts as well as grants. Both the recipients and the types of activities contracted by NASA are diverse; the latter involves most of the categories of manufacturing or service industries contained within the IMPLAN model.

NASA provided M2M-specific procurement data, listing suppliers and corresponding dollar amounts for the campaign's contractors. In addition to the geographical location of each of these suppliers including overseas vendors, the items purchased, and services provided by these vendors were detailed. Similar to total NASA procurements, the procurement activities attributable to the M2M campaign are diverse in terms of the types of economic activities and vendor locations. The M2M-specific procurement activities originating from NASA is related mostly to scientific research and development services, facilities support services, professional and technical services, software development, and electronic and precision equipment repair and maintenance.

NASA provided climate change research and technology-specific procurement data, listing suppliers and corresponding dollar amounts for the contractors. In addition to the geographical location of each of these suppliers including overseas vendors, the items purchased, and services provided by these vendors were detailed. Climate change research and technology-specific procurement activities originating from NASA are related mostly to scientific research and development services, professional and technical services, and grants and contributions.

All data and outputs of this economic impact analysis pertain to Fiscal Year 2023 (October 1, 2022, through September 30, 2023).

NASA Activities

NASA Employment

NASA had 19,758 civil servants on its payroll (17,824 FTEs, see Box 5).[18] Most of the NASA labor force is concentrated in Maryland, Texas, Virginia, Alabama, Florida, California, and Ohio (Table

[18] Throughout this report, NASA employment refers only to federal civil servants, and does not include on-site contractors. It is also worth noting that the civil servant figure represents the number of employees who NASA

1 and Figure 2). According to full-time equivalent (FTE) counts, those seven states account for 91% of total NASA employment. Because impact estimates are based on expenditures, it is important to examine the labor income shares that correspond to the employment figures in each region.[19] The same set of states accounts for the largest portion of labor income earned. As we present in the results (Section 6), the concentration of NASA employment and labor income in the states listed above translates into greater estimated household spending and larger economic impacts relative to the other regions examined.

Box 5: Civil Servant Employee Counts versus Full-time Equivalents (FTEs)

We used FTEs instead of the count of civil servants in the analysis for two reasons. First, given that there are different labor hours/labor income associated with different headcounts in different states, using FTEs provides a more accurate way to compare the share of NASA employment across states. Second, because economic impacts are a function of labor income, and changes in labor income are more directly related to FTEs, we used FTEs rather than headcounts. For example, it is possible for labor income to remain the same while the number of employees (civil servant headcounts) decreases. However, changes in FTEs would reflect those changes in labor income and thus will allow the analyst and the reader to more accurately assess the anticipated changes in associated economic impacts.

Table 1: NASA's Labor Force and Associated Labor Income [20]

State	Employment		Labor Income	
	Jobs (FTE[21])	%	Income ($ thousands)	%
Alabama	2,224	12.5	422,863	12.0
Alaska	0	0	0	0
Arizona	20	0.1	3,745	0.1
Arkansas	3	<0.1	499	<0.1
California	1,805	10.1	377,059	10.7
Colorado	56	0.3	10,742	0.3
Connecticut	9	<0.1	1,429	<0.1
Delaware	5	<0.1	857	<0.1
Florida	2,162	12.1	393,525	11.2
Georgia	27	0.1	3,815	0.1
Hawaii	7	<0.1	1,277	<0.1
Idaho	5	<0.1	877	<0.1

paid or collected payments from in FY 2023. For example, this figure includes individuals who retired in FY 2022 but received their lump sum payment in FY 2023. Thus, it does not exactly represent the number of civil servants directly employed in FY 2023.

[19] Unless otherwise mentioned, all monetary values correspond to 2023 dollars.

[20] Due to rounding decimals, column and row totals may not exactly equal the sums of the printed cell values. This caveat applies to all tables with numerical values throughout the report.

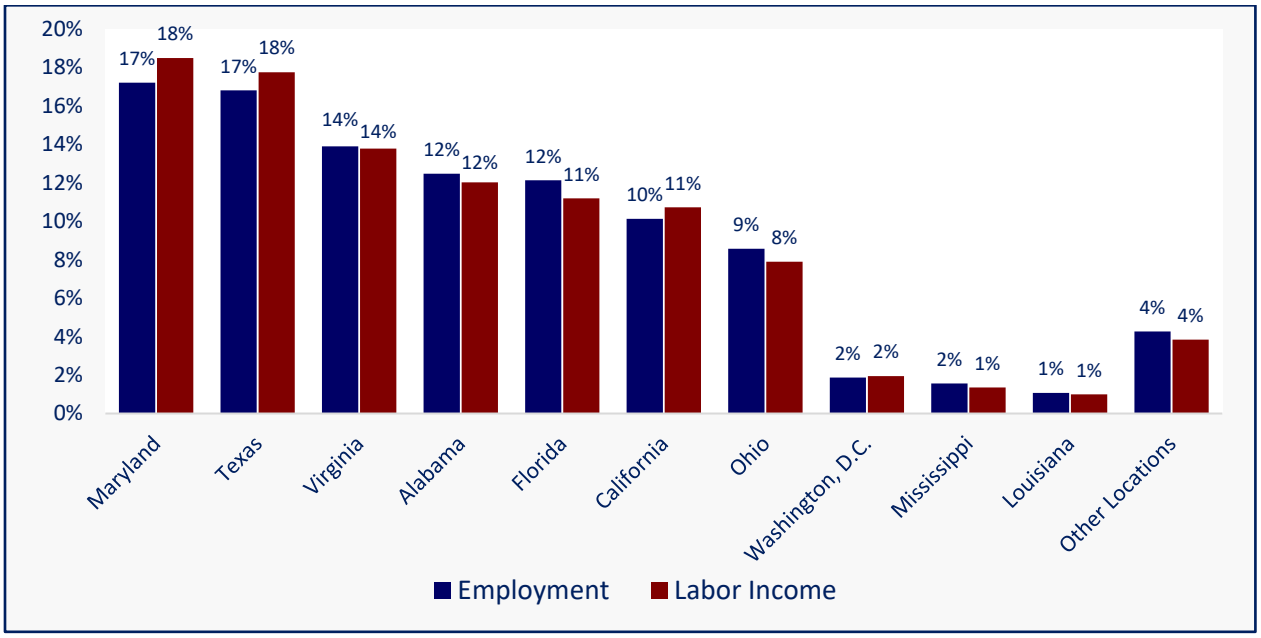
[21] The acronym FTE stands for "full-time equivalent".

Illinois	21	0.1	2,742	<0.1
Indiana	14	<0.1	2,079	<0.1
Iowa	6	<0.1	1,139	<0.1
Kansas	9	<0.1	1,356	<0.1
Kentucky	8	<0.1	1,226	<0.1
Louisiana	191	1.1	34,987	1.0
Maine	4	<0.1	657	<0.1
Maryland	3,067	17.2	650,169	18.5
Massachusetts	20	0.1	3,903	0.1
Michigan	21	0.1	2,901	<0.1
Minnesota	14	<0.1	2,051	<0.1
Mississippi	279	1.6	47,585	1.4
Missouri	13	<0.1	1,996	<0.1
Montana	3	<0.1	383	<0.1
Nebraska	2	<0.1	301	<0.1
Nevada	6	<0.1	1,016	<0.1
New Hampshire	5	<0.1	1,014	<0.1
New Jersey	23	0.1	3,895	0.1
New Mexico	85	0.5	15,422	0.4
New York	54	0.3	9,841	0.3
North Carolina	42	0.2	8,057	0.2
North Dakota	<1	<0.1	13	<0.1
Ohio	1,527	8.6	277,627	7.9
Oklahoma	13	<0.1	1,798	<0.1
Oregon	10	<0.1	1,791	<0.1
Pennsylvania	42	0.2	7,941	0.2
Rhode Island	3	<0.1	518	<0.1
South Carolina	17	<0.1	3,059	<0.1
South Dakota	4	<0.1	820	<0.1
Tennessee	71	0.4	13,514	0.4
Texas	2,997	16.8	624,309	17.8
Utah	11	<0.1	2,037	<0.1
Vermont	6	<0.1	962	<0.1
Virginia	2,478	13.9	484,604	13.8
Washington	30	0.2	5,806	0.2
Washington, D.C.	334	1.9	68,523	1.9
West Virginia	62	0.3	11,710	0.3
Wisconsin	4	<0.1	647	<0.1
Wyoming	3	<0.1	431	<0.1
Other Places	2	<0.1	414	<0.1
United States	**17,821**	**100.0**	**3,515,929**	**100.0**

U.S. Territories	2	<0.1	391	<0.1
Other Countries	0	0	0	0
Total	**17,824**	**100.0**	**3,516,320**	**100.0**

Notes: Labor income includes regular salary, over-time payments, holiday payments, OPC (Other Personnel Costs: awards, lump sum payments, bonuses, and permanent change-of-station costs), and benefits. Other places correspond to U.S. overseas military address. U.S. Territories correspond to Guam, Puerto Rico, and U.S. Virgin Islands. Values may not sum exactly due to rounding to the nearest whole number.

Figure 2: Shares of NASA Labor Force and Labor Income in Top Ten States (based on FTEs)



NASA Procurement

NASA procurement refers to contracts for goods and/or services and the P-card purchases originating from NASA headquarters and individual NASA space and flight centers. In Fiscal Year 2023, NASA incurred $23.3 billion in expenditures to vendors in 50 states, the District of Columbia, and Guam, Puerto Rico, the United States Virgin Islands, and some foreign countries. Of this amount, $7.7 billion is specific to the M2M campaign, accounting for 33.1% of total NASA procurement spending. Of the same amount (NASA procurement), $2.4 billion is specific to the investments in climate change research and technology, accounting for 10.3% of total NASA procurement spending. California is the largest recipient state, accounting for $5.8 billion, or 25% of NASA's total procurement spending.

One primary feature differentiates NASA procurement spending from NASA employment: NASA procurement expenditures are more geographically dispersed. Whereas ten states account for 96% of NASA employment, a slightly smaller share of 88% of NASA procurement is sourced from the top ten states. These different concentrations have implications for the magnitude of the

impact estimates attributable to NASA employment and procurement. (Note that on-site contractor employees are not considered directly; the impacts of these contractors are captured through procurement expenditures.)

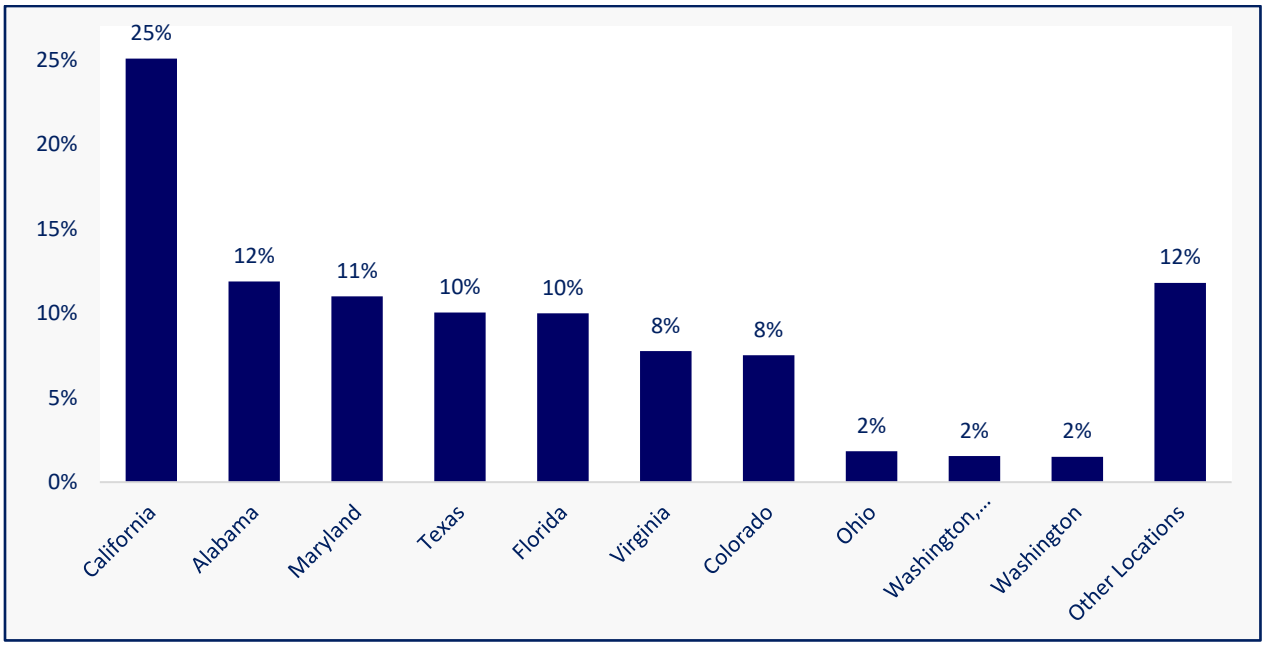
Table 2: NASA Procurement

State	Procurement		P-card Purchases		Total	
	Amount ($thousands)	%	Amount ($thousands)	%	Amount ($thousands)	%
Alabama	2,769,570	11.9	2,586	2.6	2,772,156	11.9
Alaska	23,519	0.1	17	<0.1	23,535	0.1
Arizona	221,834	1.0	2,382	2.4	224,216	1.0
Arkansas	2,762	<0.1	73	<0.1	2,834	<0.1
California	5,837,122	25.1	12,619	12.9	5,849,741	25.1
Colorado	1,750,155	7.5	2,231	2.3	1,752,386	7.5
Connecticut	56,193	0.2	1,339	1.4	57,532	0.2
Delaware	34,231	0.1	352	0.4	34,582	0.1
Florida	2,328,313	10.0	2,905	3.0	2,331,218	10.0
Georgia	27,324	0.1	1,159	1.2	28,484	0.1
Hawaii	30,377	0.1	40	<0.1	30,417	0.1
Idaho	5,333	<0.1	76	<0.1	5,409	<0.1
Illinois	53,343	0.2	9,248	9.5	62,591	0.3
Indiana	80,950	0.3	652	0.7	81,602	0.3
Iowa	52,244	0.2	92	<0.1	52,336	0.2
Kansas	11,266	<0.1	245	0.3	11,511	<0.1
Kentucky	6,673	<0.1	227	0.2	6,900	<0.1
Louisiana	147,016	0.6	216	0.2	147,233	0.6
Maine	17,025	<0.1	41	<0.1	17,066	<0.1
Maryland	2,561,718	11.0	4,341	4.4	2,566,060	11.0
Massachusetts	256,731	1.1	3,781	3.9	260,513	1.1
Michigan	62,192	0.3	1,193	1.2	63,385	0.3
Minnesota	15,648	<0.1	4,129	4.2	19,777	<0.1
Mississippi	303,193	1.3	134	0.1	303,327	1.3
Missouri	27,268	0.1	932	1.0	28,200	0.1
Montana	12,320	<0.1	165	0.2	12,485	<0.1
Nebraska	2,480	<0.1	110	0.1	2,590	<0.1
Nevada	47,902	0.2	253	0.3	48,155	0.2
New Hampshire	61,350	0.3	932	1.0	62,282	0.3
New Jersey	45,673	0.2	3,574	3.7	49,248	0.2
New Mexico	145,988	0.6	535	0.5	146,523	0.6
New York	176,694	0.8	6,249	6.4	182,942	0.8
North Carolina	61,982	0.3	1,431	1.5	63,413	0.3

North Dakota	1,470	<0.1	44	<0.1	1,514	<0.1
Ohio	423,843	1.8	3,727	3.8	427,570	1.8
Oklahoma	42,057	0.2	96	<0.1	42,153	0.2
Oregon	17,773	<0.1	789	0.8	18,563	<0.1
Pennsylvania	128,479	0.6	3,120	3.2	131,600	0.6
Rhode Island	7,378	<0.1	338	0.3	7,715	<0.1
South Carolina	10,298	<0.1	421	0.4	10,719	<0.1
South Dakota	21,544	<0.1	243	0.2	21,787	<0.1
Tennessee	63,878	0.3	1,272	1.3	65,151	0.3
Texas	2,335,271	10.1	7,448	7.6	2,342,718	10.0
Utah	159,112	0.7	1,002	1.0	160,113	0.7
Vermont	1,876	<0.1	230	0.2	2,106	<0.1
Virginia	1,803,168	7.8	7,541	7.7	1,810,709	7.8
Washington	348,564	1.5	1,527	1.6	350,091	1.5
Washington, D.C.	359,071	1.5	2,168	2.2	361,239	1.5
West Virginia	49,223	0.2	126	0.1	49,350	0.2
Wisconsin	25,454	0.1	1,222	1.3	26,677	0.1
Wyoming	1,414	<0.1	46	<0.1	1,459	<0.1
Other Places	0	0	0	0	0	0
United States	**23,036,262**	**99.2**	**95,619**	**97.9**	**23,131,881**	**99.2**
U.S. Territories	6,547	<0.1	0	0	6,547	<0.1
Other Countries	175,343	0.8	2,066	2.1	177,409	0.8
Total	**23,218,152**	**100.0**	**97,685**	**100.0**	**23,315,837**	**100.0**

Note: Other places correspond to U.S. overseas military address. U.S. Territories correspond to Guam, Puerto Rico, and U.S. Virgin Islands. Values may not sum exactly due to rounding to the nearest whole number.

Figure 3: Share of NASA Procurement Spending by Top Ten States



Moon-to-Mars (M2M) Campaign

NASA Employment for the M2M Campaign

NASA had 6,649 civil servants on its payroll (3,749 FTEs) to support the M2M campaign.[22] The majority of employees who work on the campaign reside in Texas, Alabama, and Florida; these three states account for 70% of the FTE jobs for the M2M campaign. Relatively small but notable numbers of employees live in Ohio, Virginia, Maryland, and California - other states' shares of the M2M labor force are minimal (Table 3 and Figure 4).

Table 3: NASA's M2M Campaign Labor Force and Associated Labor Income

State	Employment		Labor Income	
	Jobs (FTE)	%	Income ($ thousands)	%
Alabama	939	25.0	171,980	24.1
Alaska	0	0	0	0
Arizona	<1	<0.1	30	<0.1
Arkansas	0	0	0	0

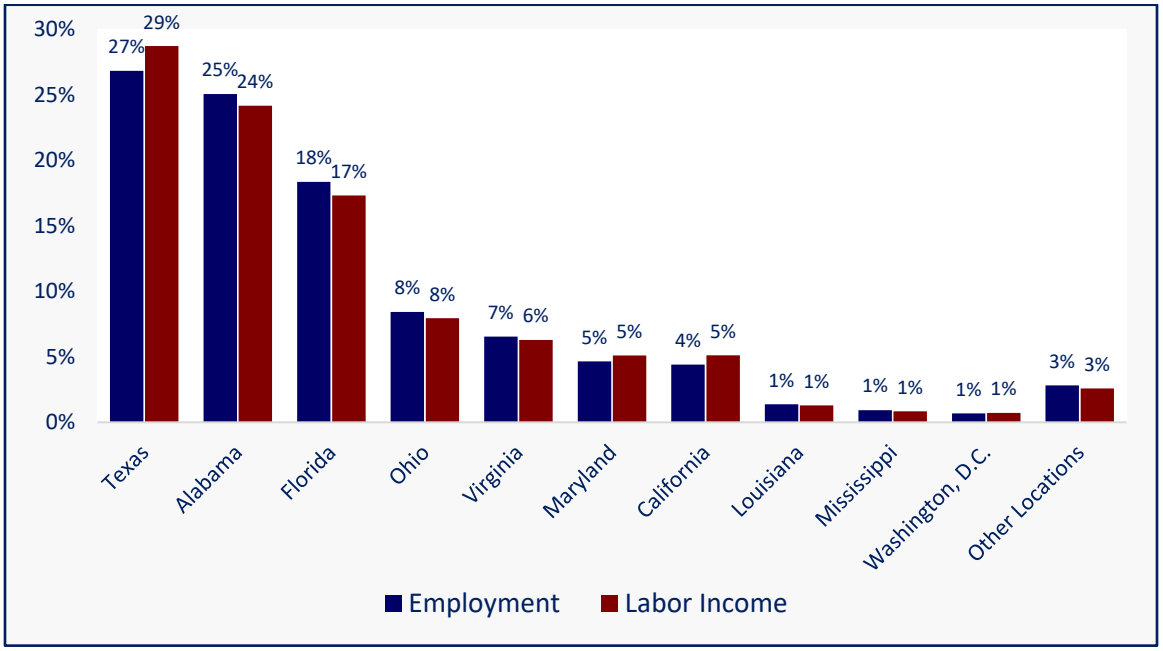
[22] Throughout this report, the M2M campaign-specific labor force figures refer only to federal civil servants, and do not include on-site contractors. It is also worth noting that the civil servant figure represents the number of employees who NASA paid or collected payments from in FY 2023. For example, this figure includes individuals who retired in FY 2022 but received their lump sum payment in FY 2023. Thus, it does not exactly represent the number of civil servants directly employed in FY 2023.

California	165	4.4	36,403	5.1
Colorado	10	0.3	1,994	0.3
Connecticut	1	<0.1	135	<0.1
Delaware	<1	<0.1	41	<0.1
Florida	688	18.3	123,197	17.3
Georgia	3	<0.1	518	<0.1
Hawaii	1	<0.1	279	<0.1
Idaho	<1	<0.1	142	<0.1
Illinois	3	<0.1	408	<0.1
Indiana	3	<0.1	386	<0.1
Iowa	<1	<0.1	16	<0.1
Kansas	1	<0.1	164	<0.1
Kentucky	3	<0.1	374	<0.1
Louisiana	52	1.4	9,148	1.3
Maine	1	<0.1	236	<0.1
Maryland	174	4.6	36,291	5.1
Massachusetts	5	0.1	951	0.1
Michigan	5	0.1	640	<0.1
Minnesota	4	<0.1	592	<0.1
Mississippi	35	0.9	6,007	0.8
Missouri	2	<0.1	344	<0.1
Montana	0	0	0	0
Nebraska	1	<0.1	119	<0.1
Nevada	<1	<0.1	10	<0.1
New Hampshire	1	<0.1	99	<0.1
New Jersey	3	<0.1	455	<0.1
New Mexico	9	0.2	1,544	0.2
New York	3	<0.1	443	<0.1
North Carolina	6	0.2	1,087	0.2
North Dakota	<1	<0.1	13	<0.1
Ohio	316	8.4	56,516	7.9
Oklahoma	1	<0.1	188	<0.1
Oregon	1	<0.1	124	<0.1
Pennsylvania	2	<0.1	361	<0.1
Rhode Island	<1	<0.1	6	<0.1
South Carolina	1	<0.1	212	<0.1
South Dakota	1	<0.1	169	<0.1
Tennessee	20	0.5	3,841	0.5
Texas	1,005	26.8	204,390	28.7
Utah	5	0.1	933	0.1
Vermont	1	<0.1	81	<0.1

Virginia	245	6.5	44,750	6.3
Washington	4	<0.1	510	<0.1
Washington, D.C.	25	0.7	5,098	0.7
West Virginia	2	<0.1	231	<0.1
Wisconsin	1	<0.1	274	<0.1
Wyoming	1	<0.1	207	<0.1
Other Places	1	<0.1	202	<0.1
United States	**3,749**	**100.0**	**712,139**	**100.0**
U.S. Territories	0	0	0	0
Other Countries	0	0	0	0
Total	**3,749**	**100.0**	**712,139**	**100.0**

Note: Labor income includes regular salary, over-time payments, holiday payments, OPC (Other Personnel Costs: awards, lump sum payments, bonuses, and permanent change-of-station costs), and benefits. Other places correspond to U.S. overseas military address. U.S. Territories correspond to Guam, Puerto Rico, and U.S. Virgin Islands. Values may not sum exactly due to rounding to the nearest whole number.

Figure 4: Shares of NASA M2M Labor Force and Labor Income in Top Ten States (based on FTEs)



NASA Procurement for the M2M Campaign

M2M procurement refers to contracts for goods and/or services and the P-card purchases originating from the different NASA Centers that support the M2M campaign**.** In Fiscal Year 2023, NASA incurred $7.7 billion in expenditures to vendors in 49 states, the District of Columbia, and